INFORMATION THEORY & CODING

Lecture 8

Review Summary

- **Entropy rate.** Two definitions of entropy rate for a *stochastic process* are

$$\begin{aligned} H(\mathcal{X}) &= \lim_{n\to\infty} \frac{1}{n} H(X_1, X_2, \ldots, X_n), \\ H'(\mathcal{X}) &= \lim_{n\to\infty} H(X_n | X_{n-1}, X_{n-2}, \ldots, X_1). \end{aligned}$$

  For a **stationary** stochastic process,

$$H(\mathcal{X}) = H'(\mathcal{X}).$$

- **Entropy rate of a stationary Markov chain.**

$$H(\mathcal{X}) = -\sum_{ij} \mu_i P_{ij} \log P_{ij}.$$

Communication system model

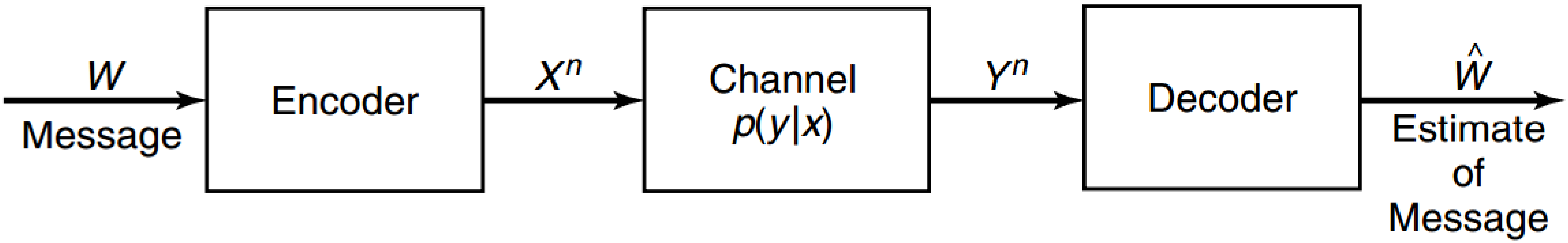


FIGURE 7.1. Communication system.

- $X^n = [X_1, X_2, \ldots, X_n]$
- $Y^n = [Y_1, Y_2, \ldots, Y_n]$
- channel: $p(y|x)$: probability of observing y given input symbol x

"Survivor"

- You were deserted on a small island. You met a native and asked about the weather.

- True weather is a random variable X

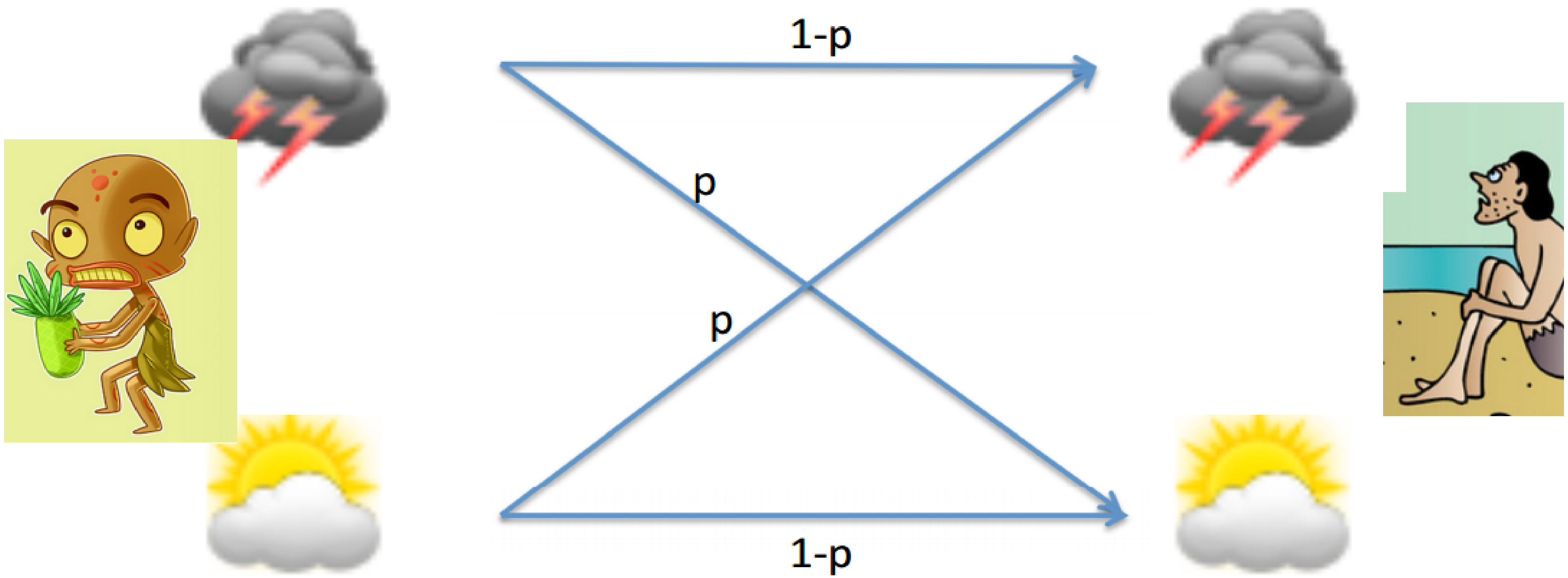
$$X = \begin{cases} \text{rain} & \text{w.p. } \alpha \\ \text{sunny} & \text{w.p. } 1-\alpha \end{cases}$$

- Native knows tomorrow's weather perfectly, but only tells truth with probability $1 - p$.

- Native's answer is a random variable $Y \in \{\text{rain}, \text{sunny}\}$.

"Survivor"

- How informative is the native's answer?



What is $I(X;Y)$?

- $I(X;Y) = H(X) - H(X|Y)$
- $H(X) = H(\alpha) = -\alpha \log \alpha - (1-\alpha)\log(1-\alpha)$
- $H(X|Y) = H(X|Y = \text{rain})p(\text{rain}) + H(X|Y = \text{sunny})p(\text{sunny})$
- $$\begin{aligned} H(X|Y=\text{rain}) &= -p(X=\text{rain}|Y=\text{rain})\log p(X=\text{rain}|Y=\text{rain}) \\ &\quad -p(X=\text{sunny}|Y=\text{rain})\log p(X=\text{sunny}|Y=\text{rain}) \end{aligned}$$

  Note that $p(X = \text{rain}|Y = \text{rain}) = \frac{p(Y=\text{rain}|X=\text{ rain})p(X=\text{rain})}{p(Y=\text{rain})} = \frac{(1-p)\alpha}{(1-p)\alpha+p(1-\alpha)}$

  Thus,

  $$H(X|Y) = \alpha H\left(\frac{(1-p)\alpha}{(1-p)\alpha+p(1-\alpha)}\right) + (1-\alpha)H\left(\frac{p\alpha}{p\alpha+(1-p)(1-\alpha)}\right)$$
- $I(X;Y) = H(\alpha) - \alpha H\left(\frac{(1-p)\alpha}{(1-p)\alpha+p(1-\alpha)}\right) - (1-\alpha)H\left(\frac{p\alpha}{p\alpha+(1-p)(1-\alpha)}\right)$

Special cases

- $I(X;Y) = H(\alpha) - \alpha H\left(\frac{(1-p)\alpha}{(1-p)\alpha + p(1-\alpha)}\right) - (1-\alpha)H\left(\frac{p\alpha}{p\alpha + (1-p)(1-\alpha)}\right)$

Special cases

- $I(X;Y) = H(\alpha) - \alpha H\left(\frac{(1-p)\alpha}{(1-p)\alpha + p(1-\alpha)}\right) - (1-\alpha) H\left(\frac{p\alpha}{p\alpha + (1-p)(1-\alpha)}\right)$

  - Always telling the truth: $p = 0$

$$I(X;Y) = H(\alpha) - \alpha H(1) - (1-\alpha)H(0) = H(\alpha) \leq 1 \text{ bit}$$

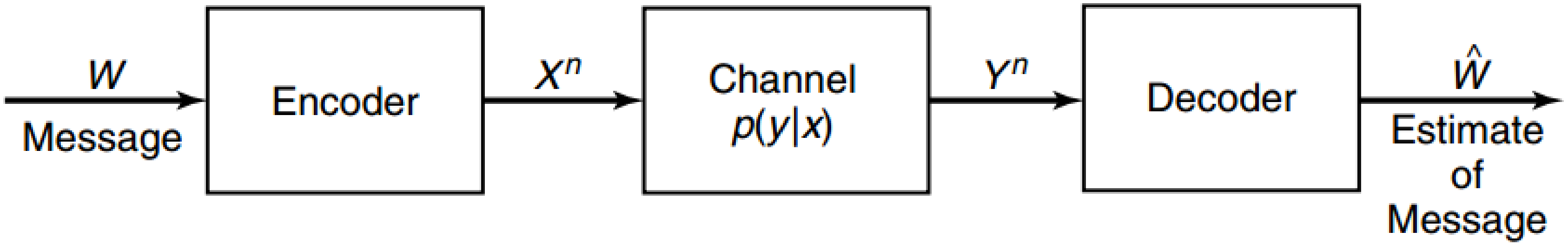
  - Telling truth half of the time: $p = 1/2$

$$I(X;Y) = H(\alpha) - \alpha H(\alpha) - (1-\alpha)H(\alpha) = 0 \text{ bit}$$

  - Fix p, maximize with respect to α, maximum achieved when $\alpha = 1/2$

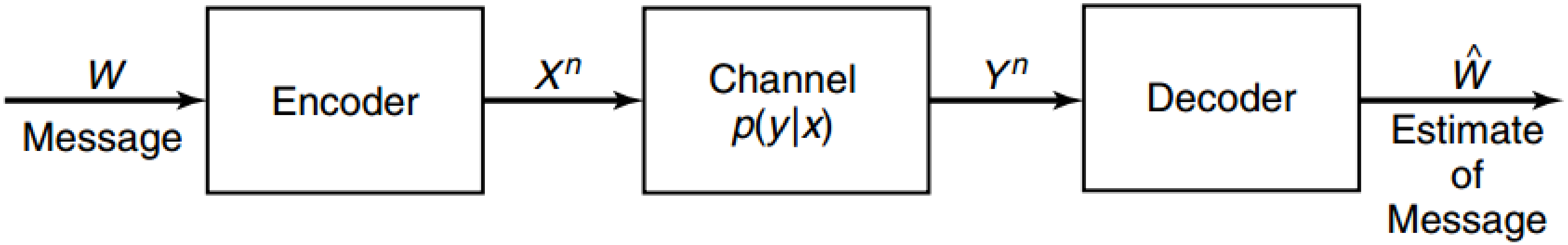
$$\max_{\alpha} I(X;Y) \leq H(1/2) - \frac{1}{2}H(1-p) - \frac{1}{2}H(p) = 1 - H(p)$$

Discrete memoryless channel (DMC)



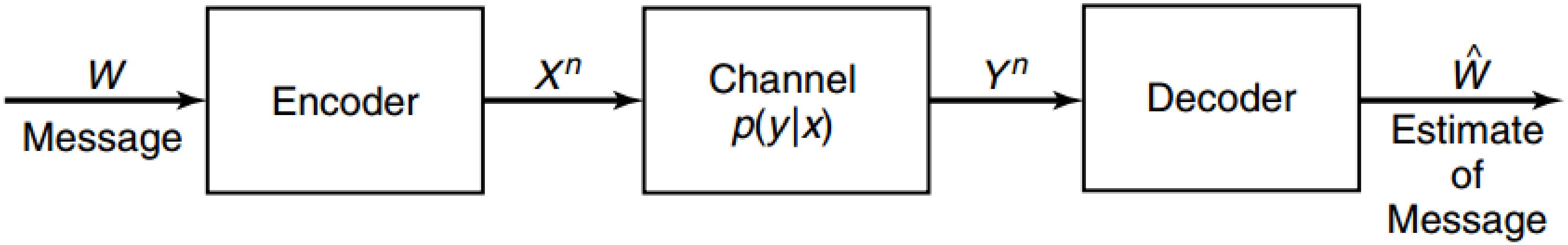
- **Definition.** A *discrete channel* consists of an input alphabet $\mathcal{X}$ and output alphabet $\mathcal{Y}$ and a probability transition matrix $p(y|x)$ that expresses the probability of observing the output symbol y given that we send the symbol x. The channel is called *memoryless* if the probability distribution of the output depends only on the input at that time and is conditionally independent of previous channel inputs or outputs.

Communication system model



- $X^n = [X_1, \ldots, X_n]$
  $Y^n = [Y_1, \ldots, Y_n]$
  channel $p(y|x)$: probability of observing y given input symbol x
  Memoryless: $p(Y^n|X^n) = \prod_{i=1}^n p(y_i|x_i)$
  $\hat{W}$: recovered message $\hat{W} = g(Y^n)$

Communication system model



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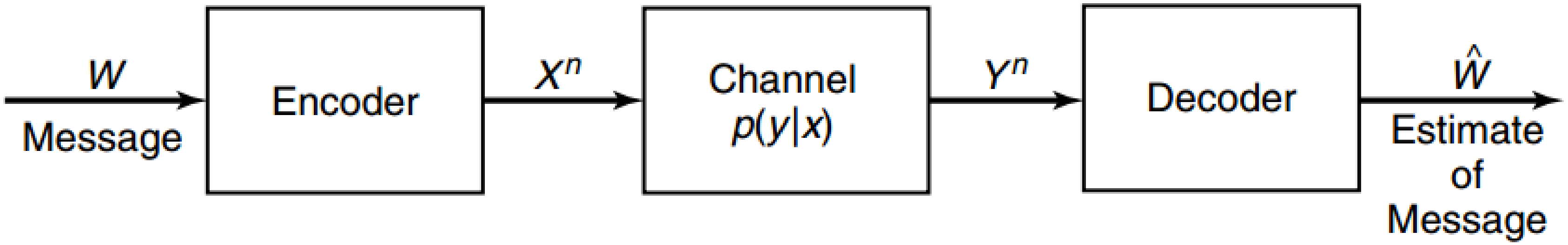
- Symbols are mapped into some sequence of the channel symbols. Output sequence is random but has a distribution that depends on the input sequences. Each possbile input sequence may induce several possbile outputs, and hence inputs are confusable. Can we choose a "non-confusable" subset of input sequences?

Duality



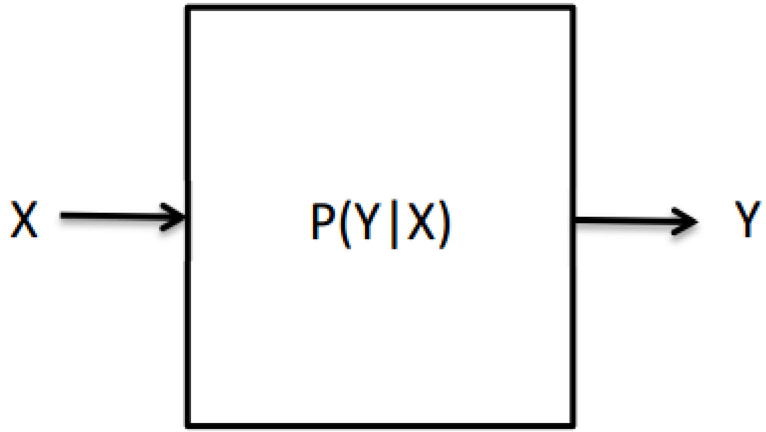
- Data compression: we remove all the redundancy in the data to form the most compressed version possible.

- Data transmission: we add redundancy in a controlled manner to combat errors in the channel.

"Information" Channel capacity

- Definition: "information" channel capacity

$$C = \max_{p(x)} I(X;Y)$$



Channel capacity: intuition

- "operational （可操作的） channel capacity"：
  $C = \log \#\{$ of identifiable inputs by passing through the channel with low error$\}$

- **Shannon's Channel Coding Theorem**

  "information channel capacity" $=$ "operational channel capacity'

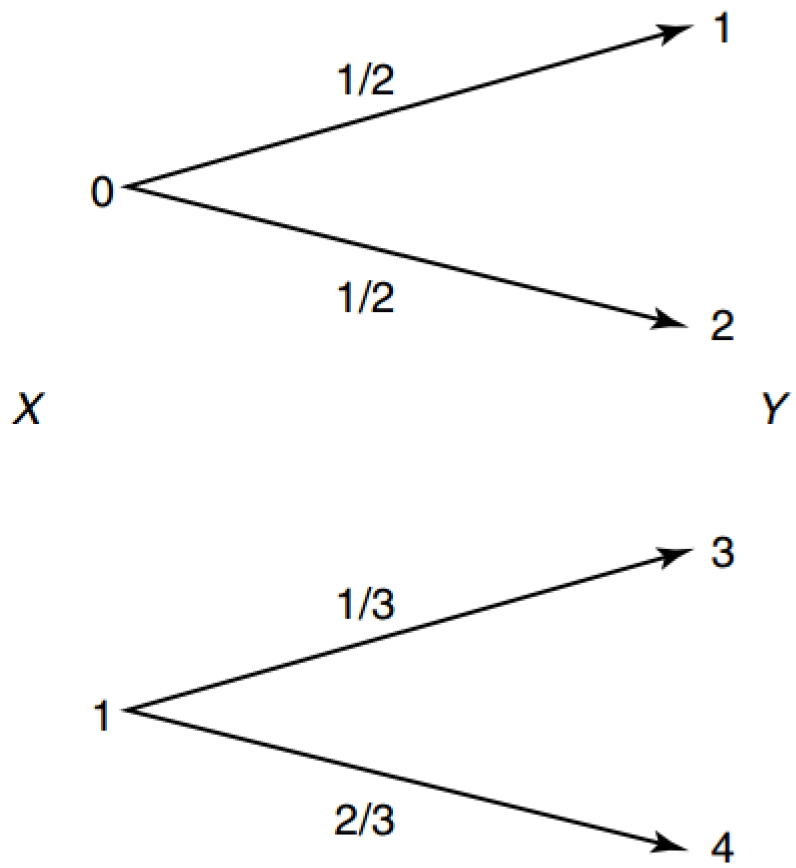
Examples

- Binary noiseless channel

$$C = \max I(X;Y) = \log 2 = 1 \text{ bit } \left(\text{ with } p(x) = \left(\tfrac{1}{2}, \tfrac{1}{2}\right) \right)$$

Examples

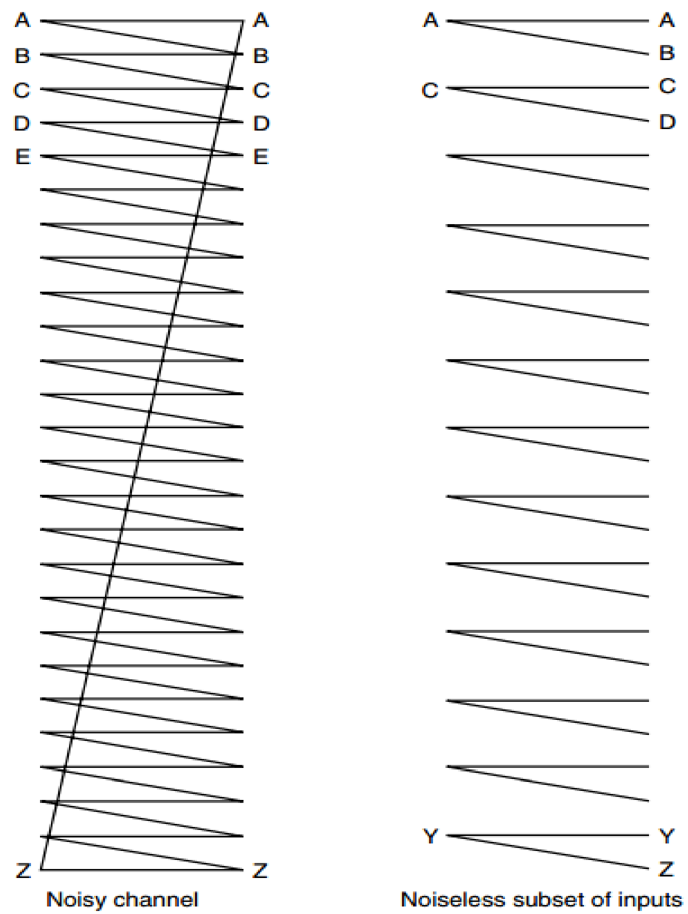
- Noisy channel with nonoverlapping outputs



$$C = \max I(X;Y) = \log 2 = 1 \text{ bit (with } p(x) = (\tfrac{1}{2}, \tfrac{1}{2}))$$

Examples

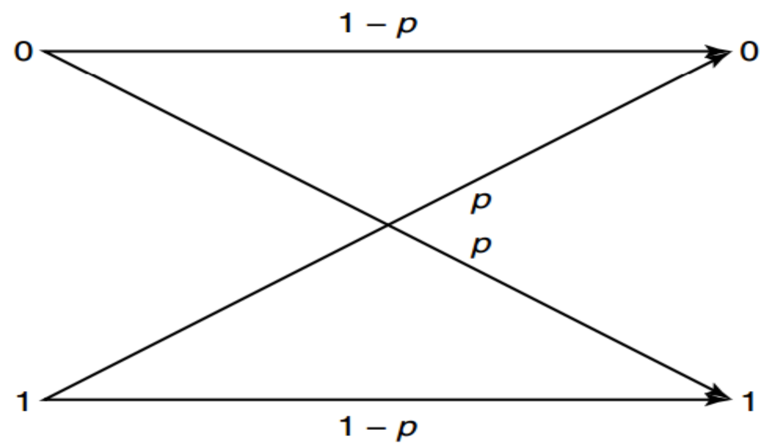
- Noisy typewriter



$C = \max I(X;Y) = \log 13$ bits (with $p(x)$ uniformly distributed)

Examples

- Binary symmetric channel

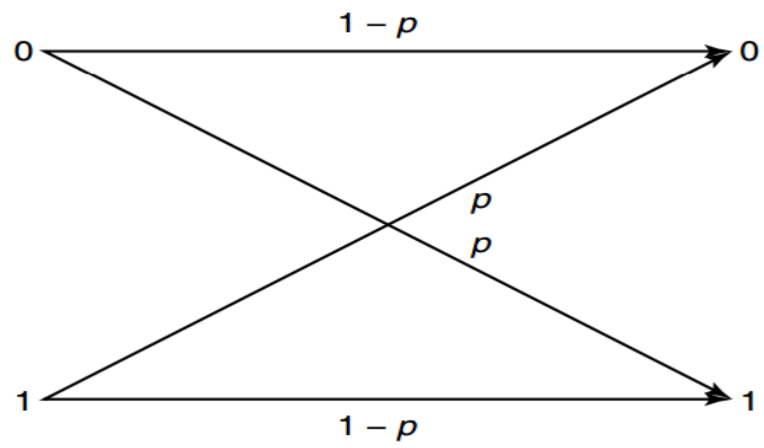


$$\begin{aligned} I(X;Y) &= H(Y) - H(Y|X) \\ &= H(Y) - \sum p(x) H(Y|X=x) \\ &= H(Y) - \sum p(x) H(p) \\ &= H(Y) - H(p) \\ &\leq 1 - H(p). \end{aligned}$$

$$C = \max I(X;Y) = 1 - H(p) \text{ bits}$$

Examples

- Binary symmetric channel



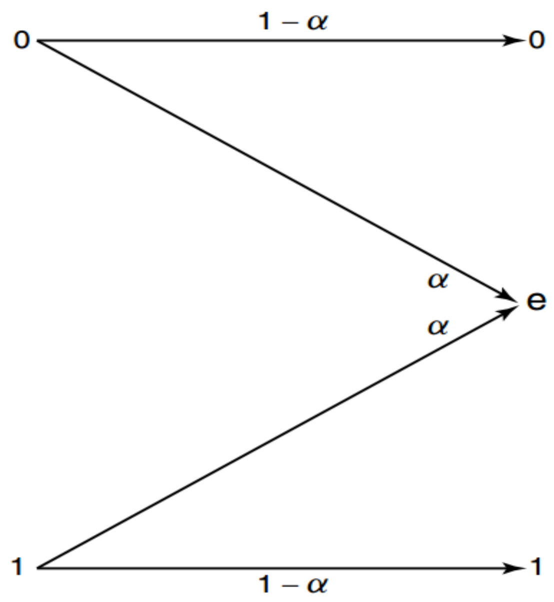
CD-ROM read channel

$$
\begin{aligned}
I(X;Y) &= H(Y) - H(Y|X) \\
&= H(Y) - \sum p(x) H(Y|X=x) \\
&= H(Y) - \sum p(x) H(p) \\
&= H(Y) - H(p) \\
&\leq 1 - H(p).
\end{aligned}
$$

$$C = \max I(X;Y) = 1 - H(p) \text{ bits}$$

Examples

- Binary erasure channel



$$
\begin{aligned}
C &= \max_{p(x)} I(X;Y) \\
&= \max_{p(x)} (H(Y) - H(Y|X)) \\
&= \max_{p(x)} H(Y) - H(\alpha)
\end{aligned}
$$

Let $\Pr[X=1] = \pi$, then

$$H(Y) = H((1-\pi)(1-\alpha), \alpha, \pi(1-\alpha)) = H(\alpha) + (1-\alpha)H(\pi).$$

Thus, $C = \max_\pi (1-\alpha)H(\pi) = 1 - \alpha$ (with $\pi = \frac{1}{2}$)

Symmetric channel

$$p(y|x) = \begin{bmatrix} 0.3 & 0.2 & 0.5 \\ 0.5 & 0.3 & 0.2 \\ 0.2 & 0.5 & 0.3 \end{bmatrix}.$$

All the rows of the transition matrix are permutations of each other and so are the columns. Let $\mathbf{r}$ be a row of the transition matrix.

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All the rows of the transition matrix are permutations of each other and so are the columns. Let $\mathbf{r}$ be a row of the transition matrix.

$$\begin{aligned} I(X;Y) &= H(Y) - H(Y|X) \\ &= H(Y) - H(\mathbf{r}) \\ &\leq \log|\mathcal{Y}| - H(\mathbf{r}) \end{aligned}$$

with equality if $\mathcal{Y}$ is uniformly distributed. If $p(x) = 1/|\mathcal{X}|$, Y is also uniformly distributed:

$$p(y) = \sum_{x\in\mathcal{X}} p(y|x)p(x) = \frac{1}{|\mathcal{X}|}\sum_{x\in\mathcal{X}} p(y|x) = \frac{c}{|\mathcal{X}|} = \frac{1}{|\mathcal{Y}|},$$

where c is the sum of the entries in one column.

Fundamental question

- How fast can we transmit information over a channel?

- Suppose a source sends r messages per second, and the entropy of a message is H bits per message, information rate is $R = rH$ bits/second.

- Intuition: as R increases, error will increase.

- Surprisingly, error can approach to zero, as long as

$$R < R_{\text{max}} = \log\{\# \text{ of distinguishable inputs}\}$$

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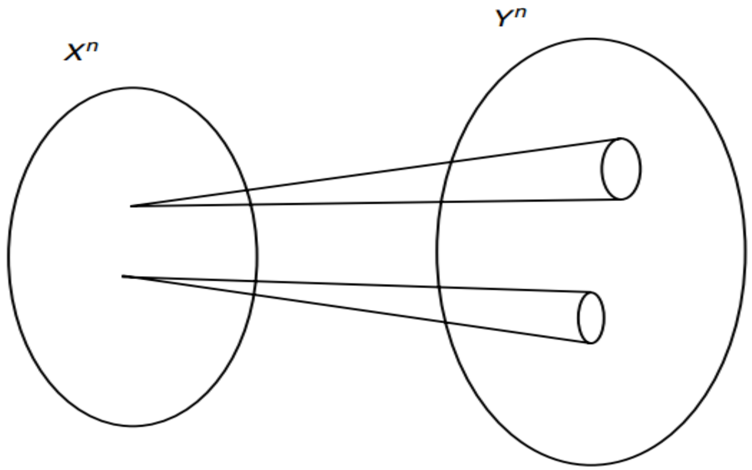
- Shannon showed $R_{\text{max}} = C$.

Basic idea



- For large block length, every channel looks like the noisy typewriter channel.
- Channel has a subset of inputs that produce "disjoint" sequences at the output.

Code rate

- **Definition.** An (M, n) code for the channel $(\mathcal{X}, p(y|x), \mathcal{Y})$ consists of:
  1. An index set $\{1, 2, \ldots, M\}$.
  2. An encoding function $X^n : \{1, 2, \ldots, M\} \to \mathcal{X}^n$, yielding codewords $x^n(1), x^n(2), \ldots, x^n(M)$. The set of codewords is called *codebook*.
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- The *rate* R of an (M, n) code is

$$R = \frac{\log M}{n} \text{ bit per transmission}$$

On the other hand, we usually write

$$M = \lceil 2^{nR} \rceil$$

Assumption about channel

- Transmit large block length: n over n transmissions

- DMC:

$$p(y^n|x^n) = \prod_{i=1}^{n} p(y_i|x_i)$$

- Channel without feedback:

$$p(x_k|x^{k-1}, y^{k-1}) = p(x_k|x^{k-1}), \quad k = 1, \dots, n$$

Performance metric

- *Conditional probability of error*:
$$\lambda_i = \Pr[g(Y^n) \neq i | X^n = x^n(i)] = \sum_{y^n} p(y^n | x^n(i)) I(g(y^n) \neq i)$$
- *Maximal probability of error*:
$$\lambda^{(n)} = \max_{i \in \{1,2,\ldots,M\}} \lambda_i$$
- *Average probability of error*:
$$P_e^{(n)} = \frac{1}{M} \sum_{i=1}^{M} \lambda_i$$
- $P_e^{(n)} \leq \lambda^{(n)}$
- If W is *uniformly distributed*:
$$P_e^{(n)} = \Pr[W \neq g(Y^n)]$$

Achievable rate

- A rate R is *achievable*,

  if there exists a sequence of $(\lceil 2^{nR} \rceil, n)$ codes such that the maximal probability of error $\lambda^{(n)} \to 0$ as $n \to \infty$. We write $(2^{nR}, n)$ codes for simplicity.

  Recall that

  The *rate* R of an (M, n) code is

$$R = \frac{\log M}{n} \quad \text{bits per tranmission.}$$

Jointly typical sequences

- The channel input X^n, output Y^n

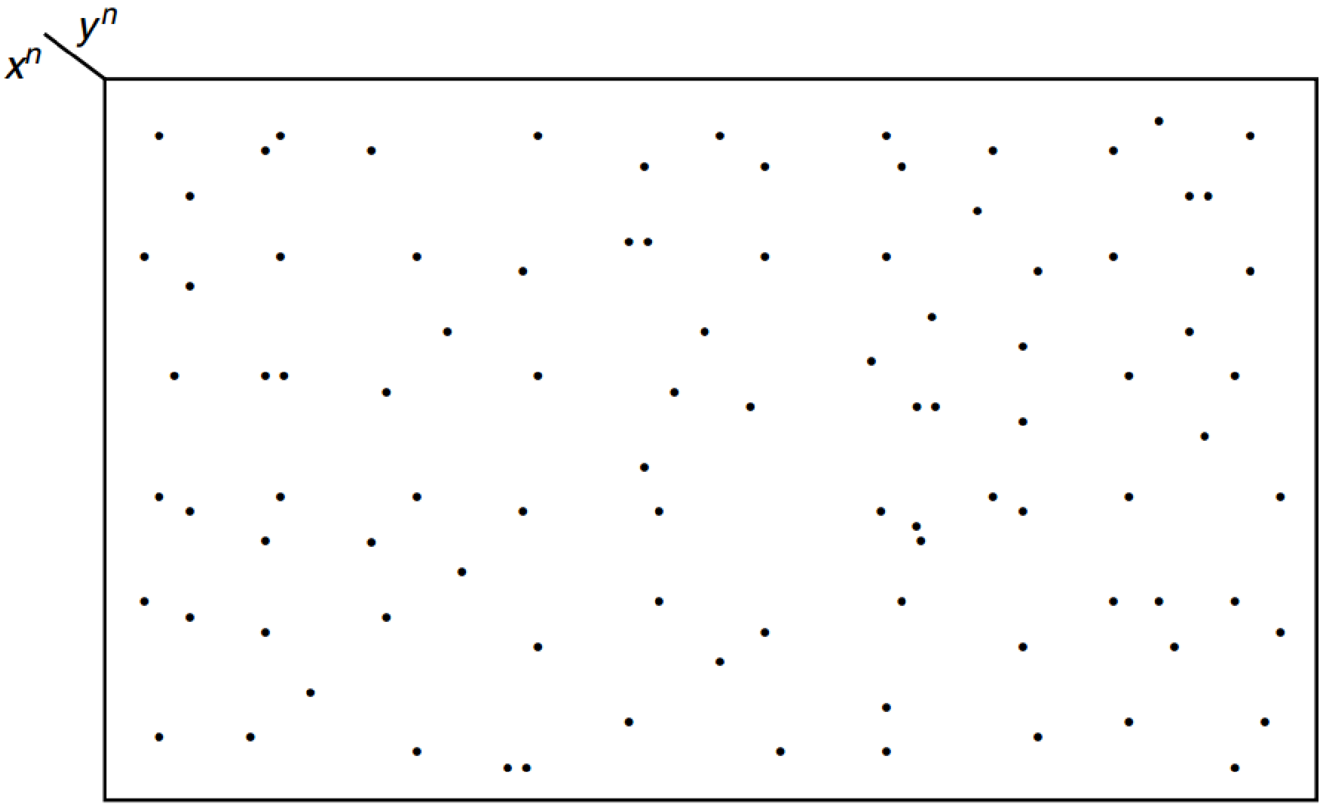
- We decode Y^n as the ith index if the codeword $X^n(i)$ is "jointly typical" with Y^n

- Set $A_\epsilon^{(n)}$ of *jointly typical sequences* $\{(x^n, y^n)\}$ is

$$
\begin{aligned}
A_\epsilon^{(n)} \quad = \quad & \{(x^n, y^n) \in \mathcal{X}^n \times \mathcal{Y}^n : \\
& \left| -\frac{1}{n} \log p(x^n) - H(X) \right| < \epsilon \\
& \left| -\frac{1}{n} \log p(y^n) - H(Y) \right| < \epsilon \\
& \left| -\frac{1}{n} \log (x^n, y^n) - H(X, Y) \right| < \epsilon \}
\end{aligned}
$$

where $p(x^n, y^n) = \prod_{i=1}^{n} p(x_i, y_i)$.

Jointly typical sequences



There are about $2^{nH(X)}$ typical X sequences and about $2^{nH(Y)}$ typical Y sequences. However, there are only $2^{nH(X,Y)}$ jointly typical sequences. The probability that any randomly chosen pair is *jointly typical* is $2^{-nI(X;Y)}$. This suggests that there are about $2^{nI(X;Y)}$ distinguishable signals X^n.

Joint AEP

- **Theorem 7.6.1** (*Joint AEP*) Let (X^n, Y^n) be sequences of length n drawn i.i.d. according to $p(x^n, y^n) = \prod_{i=1}^n p(x_i, y_i)$. Then:

  1. $\Pr[(X^n, Y^n) \in A_\epsilon^{(n)}] \to 1$ as $n \to \infty$.
  2. $|A_\epsilon^{(n)}| \leq 2^{n(H(X,Y)+\epsilon)}$.
  3. If $(\tilde{X}^n, \tilde{Y}^n) \sim p(x^n)p(y^n)$ [i.e., $\tilde{X}^n$ and $\tilde{Y}^n$ are independent with the same marginals as $p(x^n, y^n)$], then

$$\Pr[(\tilde{X}^n, \tilde{Y}^n) \in A_\epsilon^{(n)}] \leq 2^{-n(I(X;Y)-3\epsilon)}.$$

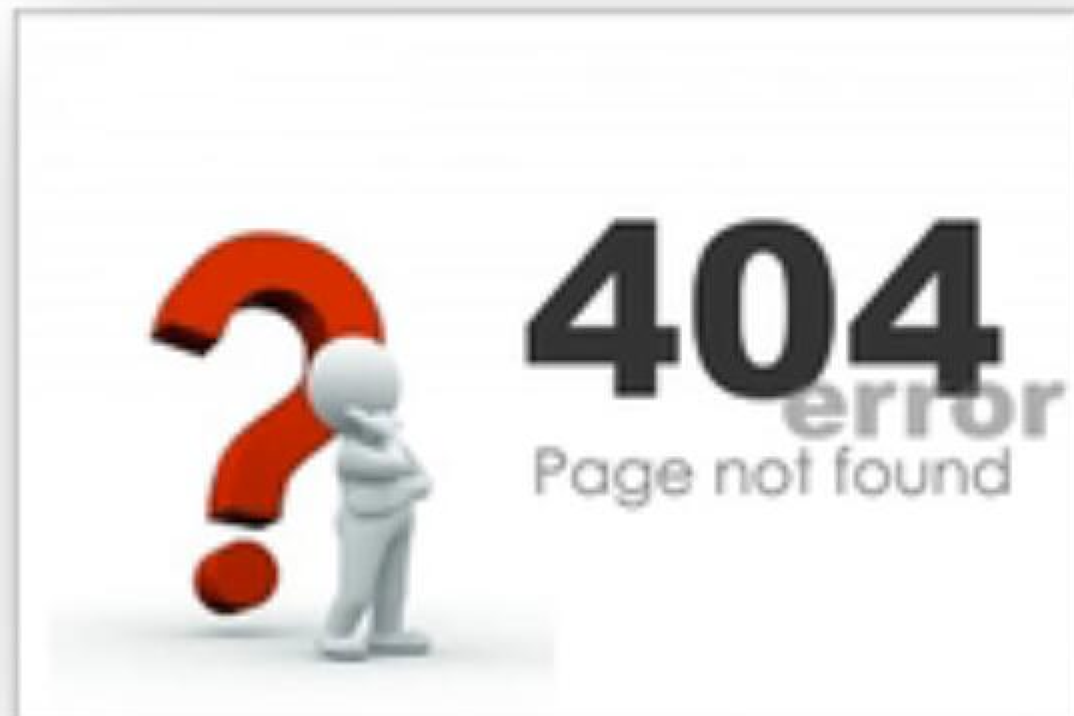
Also, for sufficently large n,

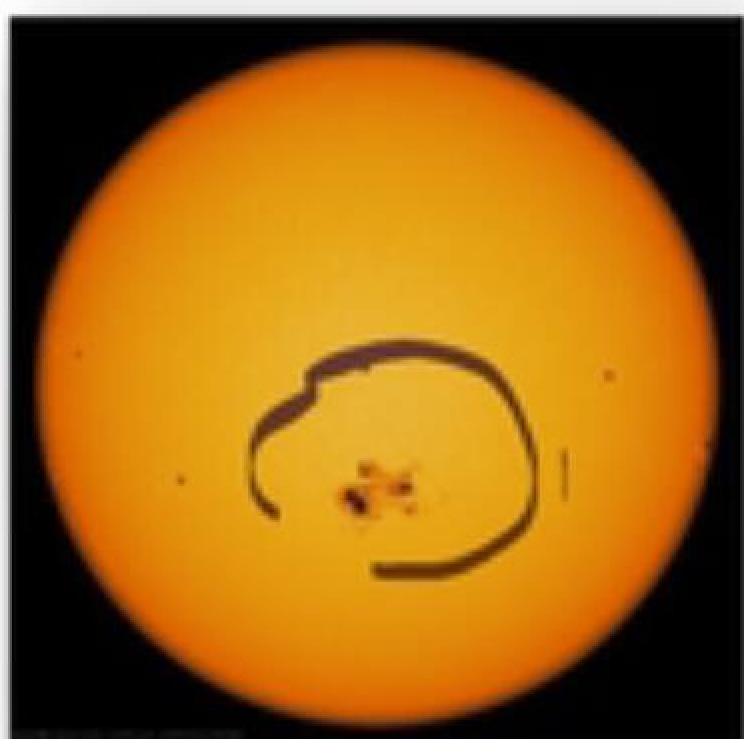
$$|A_\epsilon^{(n)}| \geq (1-\epsilon)2^{n(H(X,Y)-\epsilon)}.$$

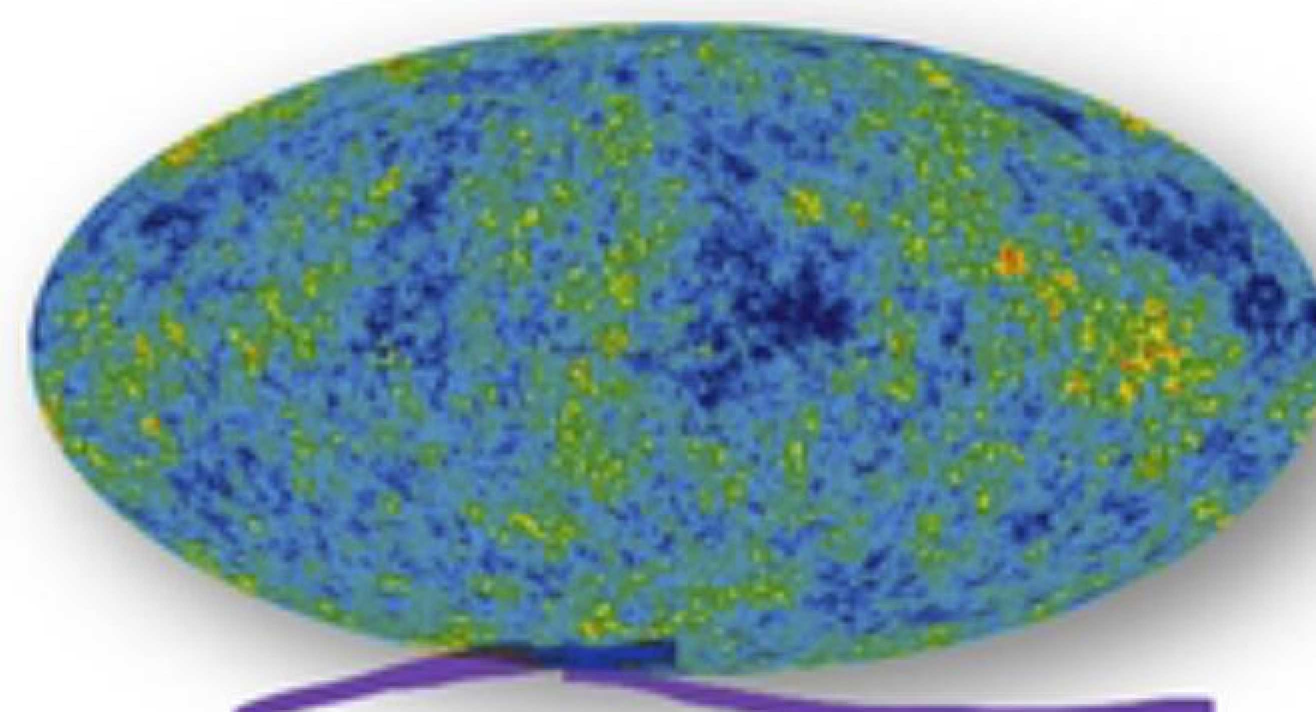
$$\Pr[(\tilde{X}^n, \tilde{Y}^n) \in A_\epsilon^{(n)}] \geq (1-\epsilon)2^{-n(I(X;Y)+3\epsilon)}.$$

INFORMATION THEORY & CODING

Noisy World



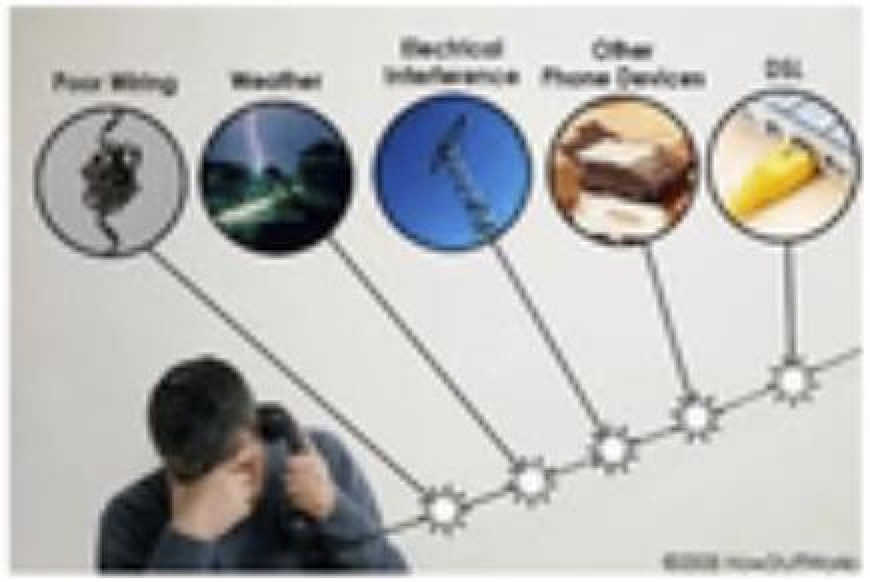




Noise cannot be eliminated from our life.
We should learn how to cope with it.

Noise in Information Transmission

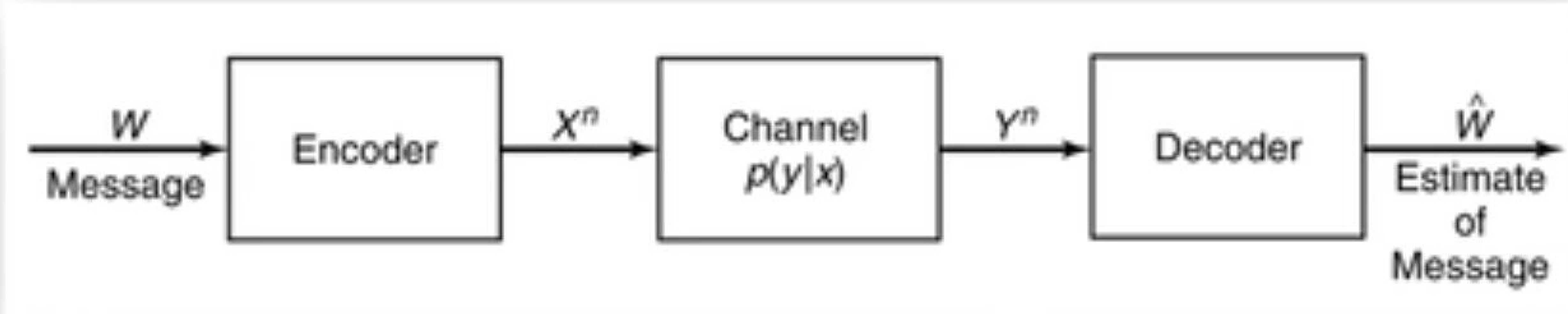
When you send your friend a message via Email/QQ/wechat, you might experience the following failures due to current network environment



- For each task, the message is M with alphabet $\mathcal{M}$
- How to model the end-to-end pipeline between the sender and the receiver
  - The input is X with alphabet $\mathcal{X}$, the output is Y with alphabet $\mathcal{Y}$. $\mathcal{X}$ and $\mathcal{Y}$ may be disjoint
  - The change from $X \rightarrow Y$ can be modeled as a transition matrix between X and Y
$$p(Y|X)$$
- The channel is just like a phone. Each time, you could use it to make a call (M)
- The message may be too large to send in just one use of the channel. Thus
$$M \rightarrow X_1, \dots, X_n$$
  That is , the channel is used n times and we use a random process $\{X_i\}$ to denote it.
- Does $\mathrm{p}(Y|X)$ remain the same for each X_i? Or we need to define $p_i(Y|X)$ for X_i

Define the right problem

Discrete Memoryless Channel



Discrete memoryless channel

- A discrete channel is a system consisting of **an input alphabet $\mathcal{X}$** and **output alphabet $\mathcal{Y}$** and **a probability transition matrix $p(y|x)$** that expresses the probability of observing the output symbol y given that we send the symbol x
- The channel is said to be **memoryless if the probability distribution of the output depends only on the input at that time and is conditionally independent of previous channel inputs or outputs**. (Each time, it is a new channel)

$(\mathcal{X}, p(y|x), \mathcal{Y})$

When you try to send x, with probability $p(y|x)$, the receiver will get y.

Channel Capacity

We define the "information" **channel capacity** of a discrete memoryless channel as

$$C = \max_{p(x)} I(X;Y),$$

where the maximum is taken over all possible input distributions $p(x)$.

- $C \geq 0$ since $I(X;Y) \geq 0$
- $C \leq \log|\mathcal{X}|$ since $C = \max I(X;Y) \leq \max H(X) = \log|\mathcal{X}|$
- $C \leq \log|\mathcal{Y}|$ for the same reason
- $I(X;Y)$ is a continuous function of $p(x)$
- $I(X;Y)$ is a concave function of $p(x)$
  - Since $I(X;Y)$ is a concave function over a closed convex set, a local maximum is a global maximum
  - $\sup I(X;Y) = \max I(X;Y)$

"$C = I(X;Y)$" **the most important formula in information age**

Properties Of Channel Capacity

General strategy to calculate C:

- $I(X;Y) = H(Y) - H(Y|X)$
  - Estimate $H(Y|X) = \sum_x H(Y|X=x)p(x)$ by the given transition probability matrix
  - Estimate $H(Y)$
- In very few situations, $I(X;Y) = H(X) - H(X|Y)$
  - Estimate $H(X|Y)$ by the given conditions in the problem
  - Estimate $H(X)$
- In general, we do not have a closed-form expression (显式表达式) for channel capacity except for some special $p(y|x)$

Example: Noiseless Binary Channel

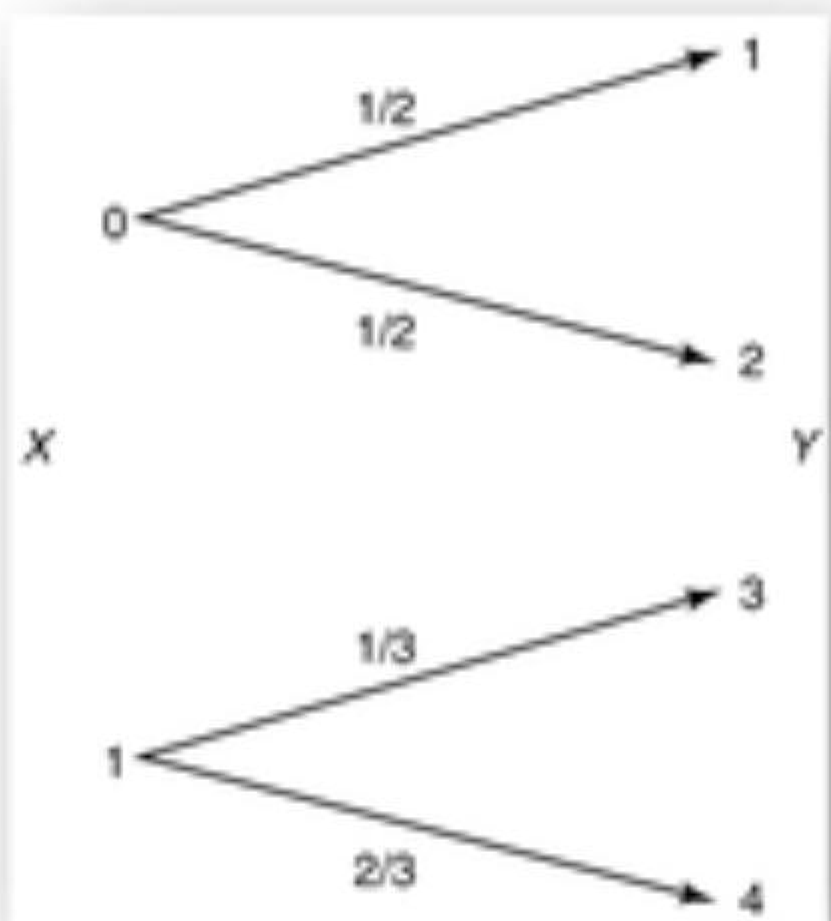
- Suppose that we have a channel whose the binary input is reproduced exactly at the output
- In this case, any transmitted bit is received without error

$$C = \max I(X;Y) = \max I(X;X) = \max H(X) \leq 1,$$

which is achieved by using $p(x) = \left(\frac{1}{2}, \frac{1}{2}\right)$.

Example: Noisy Channel with Nonoverlapping Outputs

- This channel has two possible outputs corresponding to each of the two inputs.
- The channel appears to be noisy, but really is not.

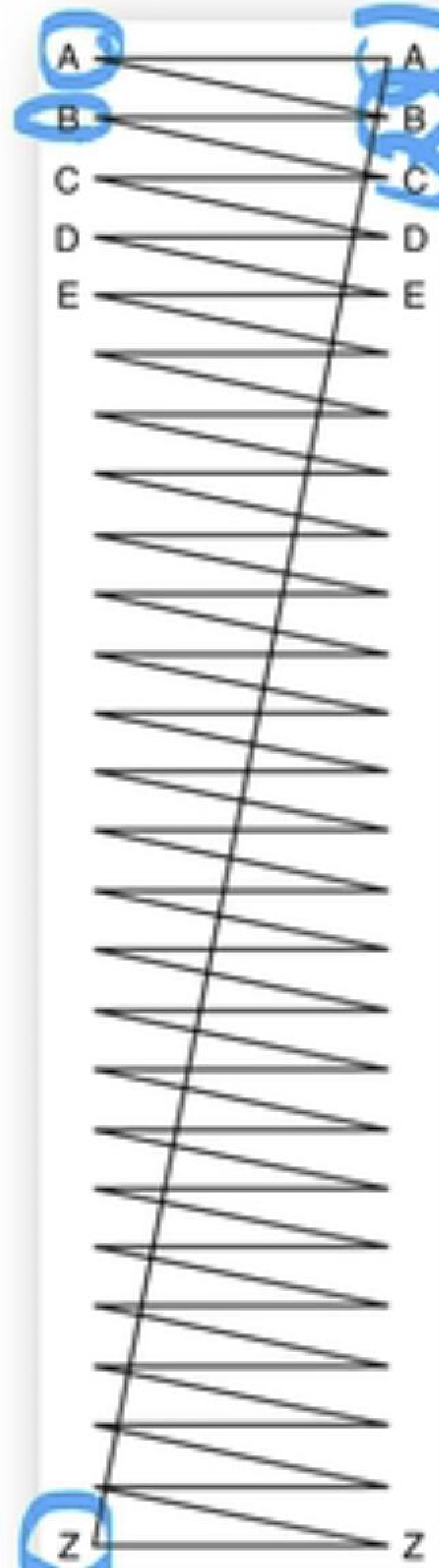


Y can determine X: X is a function of Y

Example: Noisy Typewriter

The channel input is either received unchanged at the output with probability $\frac{1}{2}$ or is transformed into the next letter with probability $\frac{1}{2}$.

$$0 \quad \frac{1}{2} \quad \frac{1}{2} \quad 0 \quad 0$$



The transition matrix: For each $x \in \{A, B, \ldots., Z\}$,

$$p(x|x) = \frac{1}{2}, \qquad p(x+1|x) = \frac{1}{2}$$

The channel looks symmetric

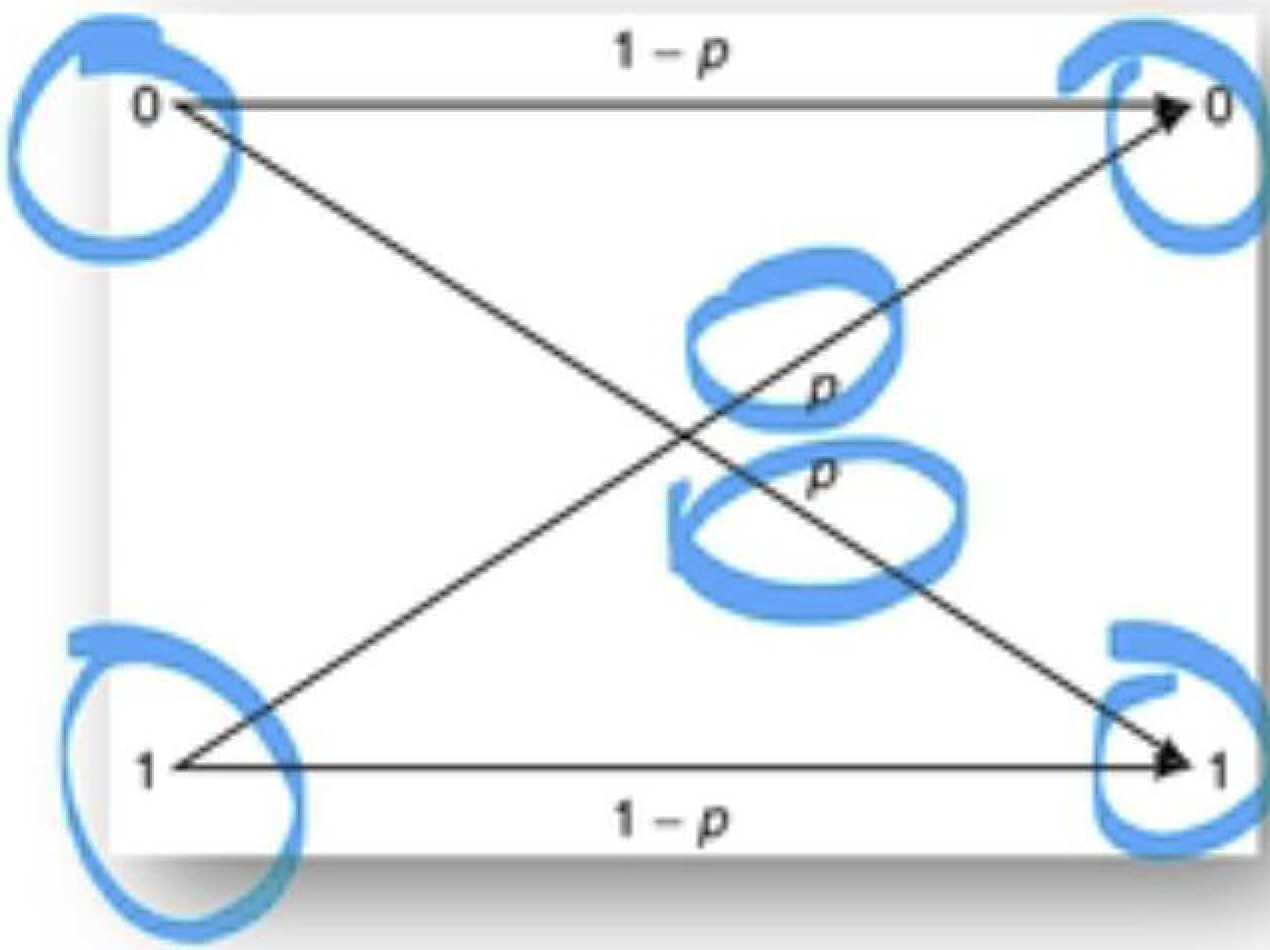
$$H(Y|X=x) = 1$$
$$H(Y|X) = \sum p(x) H(Y|X=x) = 1$$

The capacity

$$C = \max I(X;Y)$$
$$= \max(H(Y) - H(Y|X)) = \max H(Y) - 1 = \log 26 - 1 = \log 13$$
$$p(x) = \frac{1}{26}$$

Example: Binary Symmetric Channel

- When an error occurs, a 0 is received as a 1, and vice versa.



$X, Y, Z \in \{0,1\}$,

$$\Pr(Z = 0) = 1 - p$$
$$Y = X + Z \ (mod\ 2)$$
$$H(Y|X = x) = H(p)$$

$H(p)$

$$
\begin{aligned}
C &= \max I(X;Y) \\
&= \max H(Y) - H(Y|X) \\
&= \max H(Y) - \sum p(x) H(Y|X = x) \\
&= \max (Y) - \sum p(x) H(p) \\
&= \max H(Y) - H(p) \\
&\le 1 - H(p)
\end{aligned}
$$

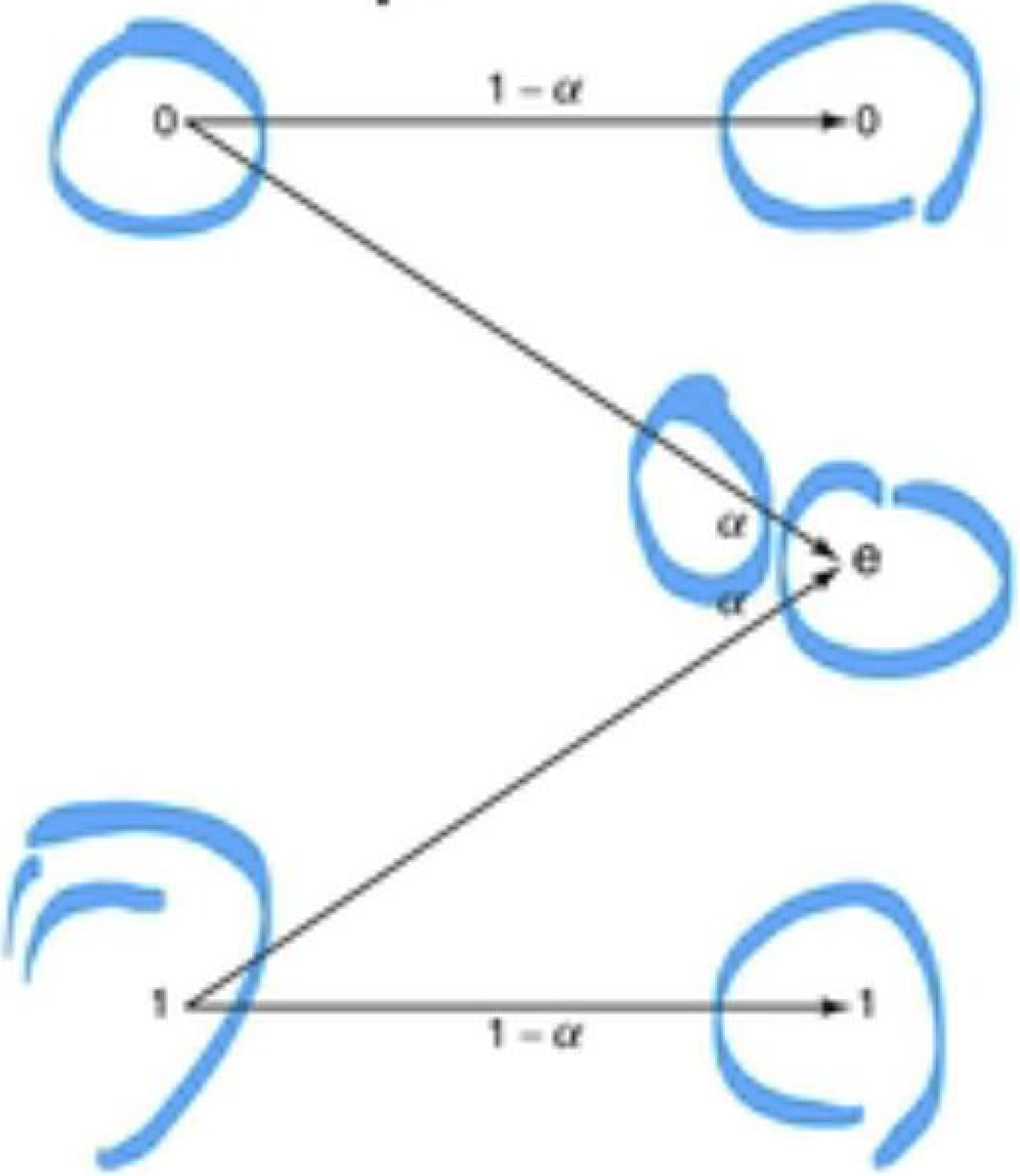
$$C = 1 - H(p)$$

$X = \frac{1}{2}$

BSC is the simplest model of a channel with errors, yet it captures most of the complexity of the general problem

Example: Binary Erasure Channel

- The analog of the binary symmetric channel in which some bits are lost (rather than corrupted) is the binary erasure channel. In this channel, a fraction α of the bits are erased.
- The receiver knows which bits have been erased. The binary erasure channel has two inputs and three outputs



$$\boldsymbol{H(Y|X=x) = H(\alpha)}$$

log3

$$\begin{aligned} C &= \max_{p(x)} I(X;Y) \\ &= \max_{p(x)} (H(Y) - H(Y|X)) \\ &= \max_{p(x)} H(Y) - H(\alpha). \end{aligned}$$

By letting $\Pr(X=1) = \pi$

$$\begin{aligned} H(Y) &= H\big((1-\pi)(1-\alpha), \alpha, \pi(1-\alpha)\big) \\ &= H(\alpha) + (1-\alpha)H(\pi) \end{aligned}$$

C$= \max_{p(x)} H(Y) - H(\alpha) = \max_{\pi}\big((1-\alpha)H(\pi) + H(\alpha) - H(\alpha)\big) =$
$\alpha)H(\pi) = 1-\alpha$

Symmetric Channel

- A channel is said to be symmetric if the rows of the channel transition matrix $p(y|x)$ are permutations of each other and the columns are permutations of each other. A channel is said to be weakly symmetric if every row of the transition matrix $p(\cdot\,|x)$ is a permutation

$$p(y|x) = \begin{bmatrix} 0.3 & 0.2 & 0.5 \\ 0.5 & 0.3 & 0.2 \\ 0.2 & 0.5 & 0.3 \end{bmatrix}, \qquad p(y|x) = \begin{bmatrix} \frac{1}{3} & \frac{1}{6} & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{2} & \frac{1}{6} \end{bmatrix}$$

Letting $\mathbf{r}$ be a row of the transition matrix, we have

$$\begin{aligned} I(X;Y) &= H(Y) - H(Y|X) \\ &= H(Y) - H(\mathbf{r}) \\ &\le \log|\mathcal{Y}| - H(\mathbf{r}) \end{aligned}$$

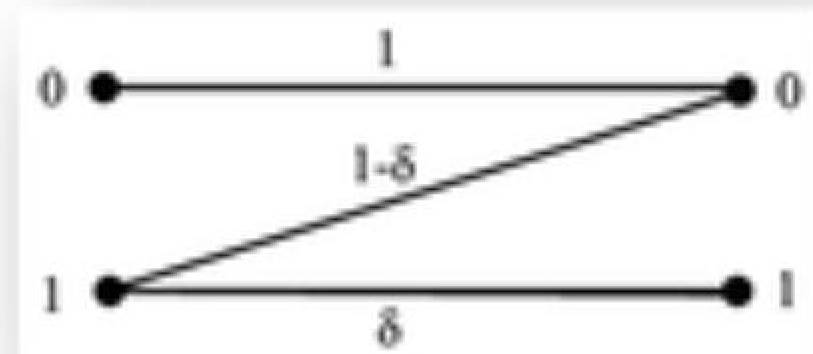
When $p(x) = \frac{1}{|\mathcal{X}|}$

$$C = \log|\mathcal{Y}| - H(\mathbf{r})$$

BSC is a special cases of symmetric channel

Exercise

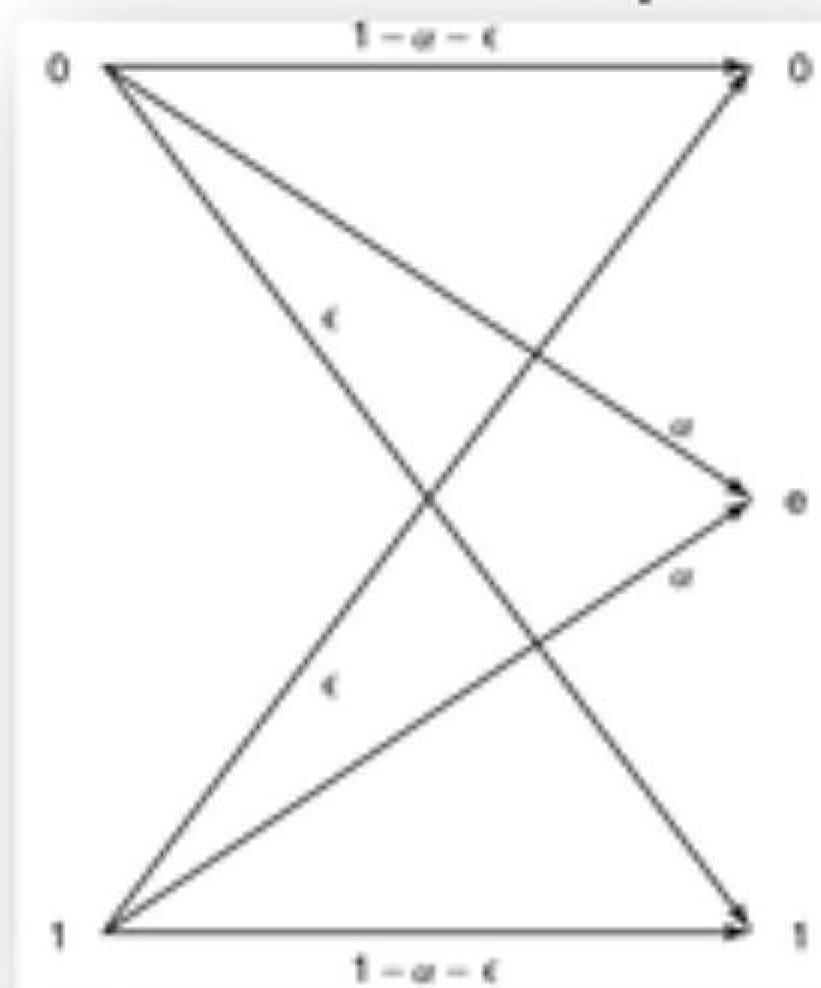
- Using two channels at once. Consider two discrete memoryless channels $(\mathcal{X}_1, p(y_1|x_1), \mathcal{Y}_1)$ and $(\mathcal{X}_2, p(y_2|x_2), \mathcal{Y}_2)$ with capacities C_1 and C_2, respectively. A new channel $(\mathcal{X}_1 \times \mathcal{X}_2, p(y_1|x_1) \times p(y_2|x_2), \mathcal{Y}_1 \times \mathcal{Y}_2)$ is formed in which $x_1 \in X_1$ and $x_2 \in X_2$ are sent simultaneously, resulting in y_1, y_2. Find the capacity of this channel.
- Z-channel. The Z-channel has binary input and output alphabets and transition probabilities $p(y|x)$ given by the following matrix:



$$Q = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}, x, y \in \{0, 1\}$$

Find the capacity of the Z-channel and the maximizing input probability distribution.

- Erasures and errors in a binary channel. Consider a channel with binary inputs that has both erasures and errors. Let the probability of error be ϵ and the probability of erasure be α, so the channel is follows:



Find the capacity of this channel.

Exercise (Cont'd)

- $\mathcal{X} = \mathcal{Y} = \{0, 1, 2\}$

$$p(y|x) = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}$$

- $\mathcal{X} = \mathcal{Y} = \{0, 1, 2\}$

$$p(y|x) = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}$$

- $\mathcal{X} = \mathcal{Y} = \{0, 1, 2, 3\}$

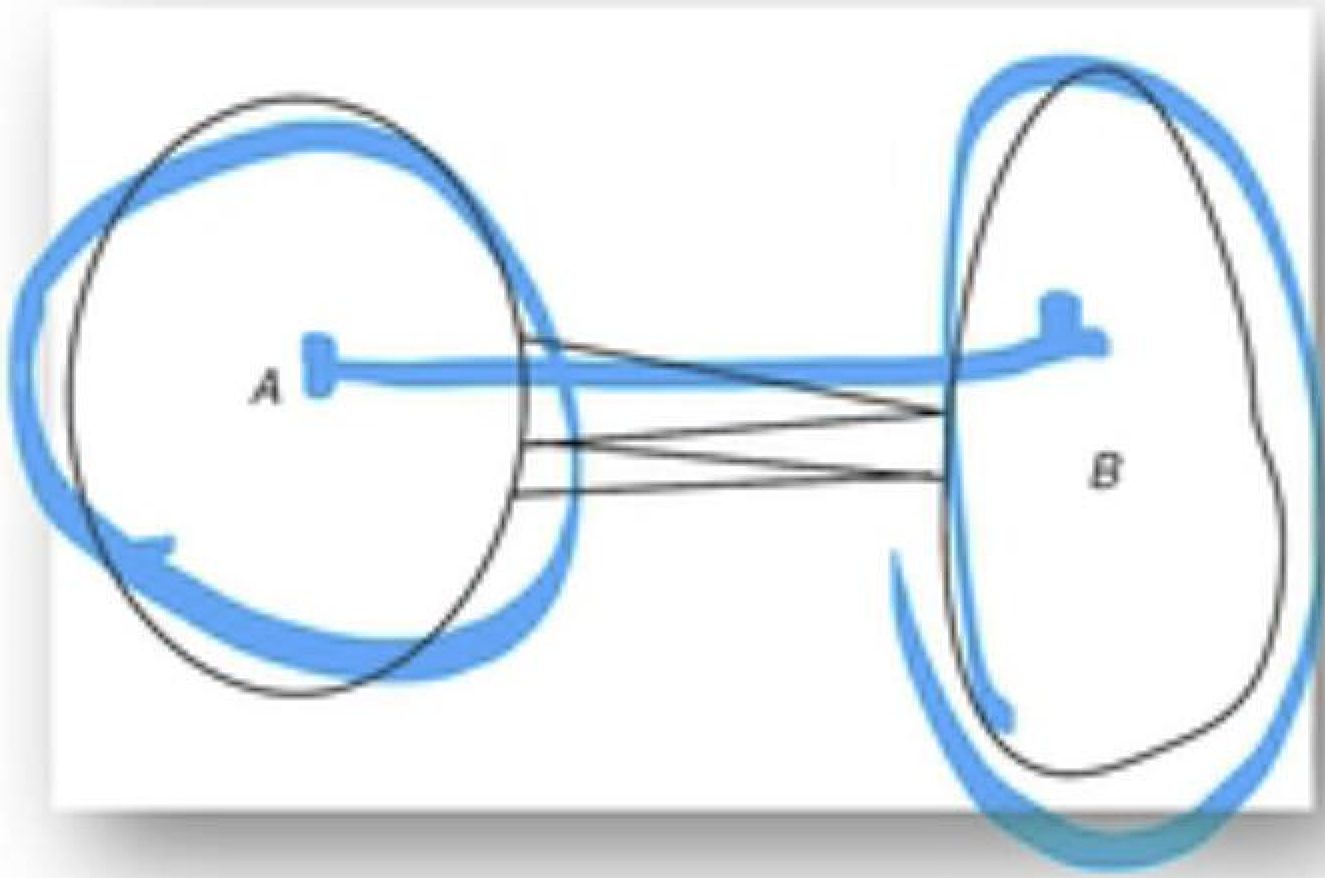
$$p(y|x) = \begin{bmatrix} p & 1-p & 0 & 0 \\ 1-p & p & 0 & 0 \\ 0 & 0 & q & 1-q \\ 0 & 0 & 1-q & q \end{bmatrix}$$

Computation Of Channel Capacity

Given **two convex sets** A and B in $\mathcal{R}^n$, we would like to find the **minimum distance** between them:

$$d_{min} = \min_{a \in A, b \in B} d(a, b)$$

where $d(a,b)$ is the Euclidean distance between a and b.



An intuitively obvious algorithm to do this would be to **take any point $x \in A$, and find the $y \in B$ that is closest to it. Then fix this y and find the closest point in A.** Repeating this process, it is clear that the distance decreases at each stage.

In particular, if the sets are sets of probability distributions and the distance measure is the **relative entropy, the algorithm does converge** to the minimum relative entropy between the two sets of distributions.

Reference: Ch. 10.8 T. Cover